

Practice Project Assignment Document

Project Title: SocialSphere – A Facebook-like Social Media Web App

Objective:

The goal of this project is to help students learn and apply core Java EE concepts using **Servlets, JSP, and JDBC** to build a **basic social media platform**. This platform will allow users to register, log in, create posts, view others' posts, and comment on them — mimicking a minimal version of Facebook's functionality.

Case Study Description

SocialSphere is a simple, text-based social media platform where users can:

1. Register themselves with a username, password, and email.
2. Log in using their credentials.
3. Create a new text post.
4. View all posts made by all users.
5. Comment on any existing post.

All data must be stored and retrieved from a relational database using **JDBC**. The user interface will be created using **plain HTML (no CSS or JavaScript)**, and the server-side logic will be built using **Servlets and JSP**.

Tech Stack

Layer	Technology Used
Frontend	HTML only (no CSS/JS)
Server-side	Java Servlets & JSP
Backend DB	H2 db
Database Access	JDBC
IDE Suggested	Eclipse / IntelliJ IDEA
Server	Apache Tomcat (v11)

Functional Requirements

1. User Registration

- Form to accept username, password, and email.
- Store these details in the database after checking for existing users.

2. User Login

- Form to accept username and password.
- Authenticate user against the database.
- Show error if credentials are incorrect.

3. Create Post

- After login, users should see a form to create a post (title and content).
- Each post should be stored in the database with the user ID and timestamp.

4. View All Posts

- Display all posts submitted by all users in reverse chronological order.
- Each post should show the username, title, content, and date/time.

5. Comment on Posts

- Under each post, show a form to submit a comment.
- Save the comment in the database with the user ID, post ID, and timestamp.
- Display existing comments under each post.

Bonus Suggestions

- Add a logout feature using session handling.
- Display post and comment counts on the user dashboard.